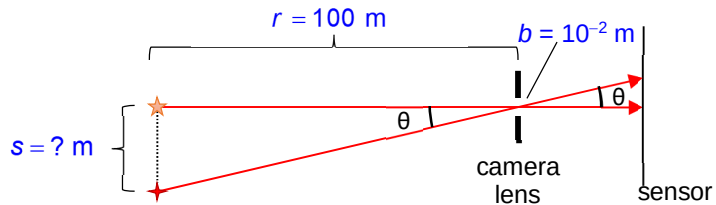


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Answer: C



From circular measure

$$s \approx r\theta$$

Rayleigh's Criterion

$$\theta \approx \frac{\lambda}{b}$$

For small angle,

$$\frac{s}{r} \approx \theta \approx \frac{\lambda}{b}$$

$$s \approx \frac{r\lambda}{b}$$

$$= \frac{(100)(550 \times 10^{-9})}{10 \times 10^{-2}}$$

$$= 5.5 \times 10^{-4} \text{ m} = 0.55 \text{ mm}$$

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Answer: D